STAT 7900: Special Topic: Causal Analysis



Instructor

Dhrubajyoti Ghosh

Kennesaw State University

STAT 7900

From ATE to CATE

$$ATE: \boldsymbol{\tau} = E[Y(1) - Y(0)] \quad \textbf{vs.} \quad CATE: \tau(x) = E[Y(1) - Y(0) \mid X = x]$$

ATE: one summary

How much does the treatment change the outcome on average across the target population?

CATE: effect heterogeneity

Does the treatment effect depend on baseline covariates such as age, risk, or previous earnings?

Key caution: CATE is an average for people with a given X, not an observed individual causal effect.

Classical HTE: The interaction-model bridge

$$E[Y \mid T, X] = \beta_0 + \beta_1 T + \beta_2 X + \beta_3 (T \times X)$$

Estimated conditional effect: $\tau(x) = \beta_1 + \beta_3 x$

What we gain

A testable, interpretable treatment-effect modifier.

What becomes hard

Many modifiers, nonlinear effects, and unknown interactions.

This motivates flexible CATE learners without manually specifying every interaction.

S-, T-, and X-learners: Three workflows

S-learner • single model

Fit $m(T, X)$ using everyone.

Predict $m(1, x)$ and $m(0, x)$; subtract.

T-learner • two models

Fit $m_1(x)$ among treated and $m_0(x)$ among controls.

Subtract the predictions.

X-learner • two stages

Start with m_1 and m_0.

Impute effects, fit effect models, and combine them.

None is universally best. All still rely on valid identification assumptions.

X-learner: Recall the pseudo-treatment effects

Among treated $(T = 1)$

D^1 = observed Y − predicted $m_0(X)$

Among controls $(T = 0)$

D^0 = predicted $m_1(X)$ − observed Y

Then fit two models of D^1 and D^0 as functions of X.

Combine: $\hat{\tau}_X(x) = e(x)\,\hat{\tau}_0(x) + [1 - e(x)]\hat{\tau}_1(x)$

Important: D^1 and D^0 are imputed effect proxies, not true observed individual effects.

Comparison: S-, T-, and X-Learners

Feature	S-Learner	T-Learner	X-Learner
Outcome models	1	2	2 initial + effect models
Information sharing	High	Low	Intermediate / adaptive
Treatment imbalance	Can be stable	Can struggle	Often attractive
Main idea	Predict both worlds from one model	Separate response surfaces	Impute and learn treatment effects
Main risk	Treatment signal shrunk	Small-arm instability	More modeling stages

None is uniformly best; performance depends on sample size, imbalance, smoothness, and the base learner.

From concepts to R: What will we implement?

S-learner

One `lm()` with treatment × covariate interactions. Predict twice per person.

T-learner

Separate `lm()` fits for the two arms.
Predict both potential-outcome means.

X-learner

Outcome models → D^1/D^0 → effect models → propensity weighting.

```
install.packages("MatchIt")
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
head(d)
```

R Code: S-Learner

Code

```
fit_s <- lm(
  re78 ~ treat * (age + educ + re74 + re75),
  data = d
)
d1 <- transform(d, treat = 1)
d0 <- transform(d, treat = 0)
mu1_s <- predict(fit_s, newdata = d1)
mu0_s <- predict(fit_s, newdata = d0)
tau_s <- mu1_s - mu0_s
summary(tau_s)
```

How to read it

1. Fit one outcome model using T and X together.
2. Create the all-treated and all-control prediction worlds.
3. For each person, subtract the two predicted outcomes.

Important: with a linear model, include $treat \times X$ if you want heterogeneity.

R Code: S-Learner

Code

```
cate_s <- function(age, educ, re74,
re75) {
  b <- coef(fit_s)

  b["treat"] +
    b["treat:age"] * age +
    b["treat:educ"] * educ +
    b["treat:re74"] * re74 +
    b["treat:re75"] * re75
}
```

How to read it

- coef(fit_s) extracts the estimated regression coefficients.
- The treatment coefficient and treatment–covariate interaction coefficients define the estimated CATE function.
- Different covariate values can produce different estimated treatment effects.

R Code: T-Learner

Code

```
fit1 <- lm(
  re78 ~ age + educ + re74 + re75,
  data = d[d$treat == 1, ]
)
fit0 <- lm(
  re78 ~ age + educ + re74 + re75,
  data = d[d$treat == 0, ]
)
mu1_t <- predict(fit1, newdata = d)
mu0_t <- predict(fit0, newdata = d)
tau_t <- mu1_t - mu0_t
summary(tau_t)
```

How to read it

1. Fit a treated-outcome model only among treated units.
2. Fit a control-outcome model only among control units.
3. Use both models to predict for the same person.
4. Subtract the two predicted outcomes.

This allows the treated and control response surfaces to differ.

R Code: X-Learner (Steps 1–2)

Code

```
# Step 1: initial outcome models
fit1 <- lm(re78 ~ age + educ + re74 + re75,
           data = d[d$treat == 1, ])
fit0 <- lm(re78 ~ age + educ + re74 + re75,
           data = d[d$treat == 0, ])

m1 <- predict(fit1, newdata = d)
m0 <- predict(fit0, newdata = d)

# Step 2: imputed treatment effects
D1 <- d$re78[d$treat == 1] - m0[d$treat == 1]
D0 <- m1[d$treat == 0] - d$re78[d$treat == 0]
```

How to read it

For treated units:

D1 = observed Y(1) − predicted Y(0)

For control units:

D0 = predicted Y(1) − observed Y(0)

These are pseudo-treatment effects.

R Code: X-Learner (Steps 3–4)

Code

```
# Step 3: model the pseudo-effects
fit_tau1 <- lm(D1 ~ age + educ + re74 + re75,
               data = d[d$treat == 1, ])
fit_tau0 <- lm(D0 ~ age + educ + re74 + re75,
               data = d[d$treat == 0, ])
tau1 <- predict(fit_tau1, newdata = d)
tau0 <- predict(fit_tau0, newdata = d)

# Step 4: combine
ps_fit <- glm(treat ~ age + educ + re74 +
re75,
              family = binomial(), data = d)
e <- predict(ps_fit, type = 'response')
tau_x <- e * tau0 + (1 - e) * tau1
summary(tau_x)
```

How to read it

1. Learn how the pseudo-effects vary with X.
2. Predict those effect models for everyone.
3. Combine the two effect-model predictions.

The propensity score helps decide how much weight to give each side.

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Why Do We Need Another HTE Method?

S / T / X learners

Learn outcome models and/or imputed effects.

Result depends on the choice and quality of base learners.

What is still difficult?

Many modifiers, nonlinear interactions, sparse subgroups, and data-driven searches for where effects differ.

A different idea

Build local treatment–control comparisons around people with similar X, using trees designed for HTE.

Causal forest is an alternative CATE estimator, not a required successor to S/T/X.

Ordinary vs Causal Trees: What Should a Split Find?

Ordinary prediction tree

Split on X to make outcomes Y easier to predict.

Example: baseline severity predicts risk well, even when treatment effect is constant.

Causal tree

Split on X when there is evidence that treatment effects differ across the resulting groups.

Example: biomarker status changes the treated-control contrast.

Strong predictor of Y $\neq$ strong modifier of treatment effect

Ordinary Random Forest Is Not a Causal Forest

Ordinary random forest

- Goal: predict Y accurately.
- Splits reduce outcome prediction error.
- A strong prognostic variable may dominate even if it does not modify treatment effect.

Causal forest

- Goal: estimate heterogeneity in treatment effects.
- Splits seek regions with different treatment effects.
- Treatment and control information are compared within leaves.

Prediction heterogeneity ≠ treatment-effect heterogeneity

Causal Forest: Intuition

Grow many trees

Repeatedly partition covariate space into local neighborhoods.

Look for effect differences

Prefer splits that reveal treatment-effect heterogeneity rather than outcome levels alone.

Average locally

Combine many trees to estimate $\tau(x)$ for each covariate profile.

Forest = adaptive local comparison + aggregation across many trees

Causal Forest: The Main Idea

1. Local groups

Trees partition covariate space into neighborhoods of similar X.

2. Local causal contrasts

Within neighborhoods, use observed treatment and outcome information to estimate effects.

3. Ensemble

Average information across many trees to produce a CATE estimate at x.

Unlike an ordinary forest, splitting is designed for treatment-effect heterogeneity.

Honest Trees: Choosing Splits

Without sample separation

The same unusual outcomes can influence both the tree partition and the effect estimate.

This can exaggerate apparent heterogeneity.

Honest estimation

Use one subsample to choose splits and another to estimate treatment effects in the selected leaves.

Reduces adaptive estimation bias.

Honesty helps with adaptive bias; it does not remove confounding or lack of overlap.

What Causal Forests Do and Do Not Guarantee

Useful features

- Learns complex interactions without listing every $T \times X$ term
- Pools information through many trees
- Produces a CATE prediction for a covariate profile
- Some implementations support statistical inference

Still required

- A defensible causal design and adjustment set
- Adequate overlap and sample size
- Valid outcome and treatment coding
- Diagnostics and honest uncertainty reporting

A sophisticated algorithm does not identify a causal effect by itself.

R: Fit a Causal Forest with the LaLonde Data

R code

```
library(MatchIt)
library(grf)
data("lalonde", package = "MatchIt")
d <- lalonde
X <- model.matrix(
  ~ age + educ + race + married + nodegree + re74 + re75,
  data = d)[, -1, drop = FALSE]

cf <- causal_forest(
  X, Y = d$re78, W = d$treat, seed = 123
)
```

X: baseline covariate matrix

Y: 1978 earnings

W: training indicator

The forest estimates effects conditionally on X.

Requires package grf.

R: Fit a Causal Forest with the LaLonde Data

R code

```
cate_cf <- predict(cf)$predictions
summary(cate_cf)
new_person <- d[1, , drop = FALSE]
new_person$age <- 30
new_person$educ <- 12
new_person$re74 <- 5000
new_person$re75 <- 6000
# Build the same predictor matrix
X_new <- model.matrix(
  ~ age + educ + race + married + nodegree + re74 + re75,
  data = new_person
)[, -1, drop = FALSE]

predict(cf, newdata = X_new)$predictions
```

How to use the fitted forest to get cate estimates

How to see how CATE changes with age?

R code

```
# Use a typical worker as reference
X_age <- X[rep(1, 38), , drop = FALSE]
# Change only age
ages <- 18:55
X_age[, "age"] <- ages
# Predict CATE at each age
tau_age <- predict(cf, newdata = X_age)$predictions
```

```
plot(
  ages, tau_age,
  type = "l",
  lwd = 2,
  xlab = "Age",
  ylab = "Estimated CATE ($)",
  main = "Causal Forest: Training Benefit by
Age"
)

abline(h = 0, lty = 2)
```

What Does Variation in Predicted CATEs Mean?

Possible signal

The causal effect really differs across baseline covariate profiles.

Possible noise

Small subgroups, model instability, or finite-sample variability create apparent differences.

Needed next

Calibration, uncertainty, external or honest validation, and overlap checks.

Prediction variability is a hypothesis about HTE, not a completed validation.

Can We Trust the Heterogeneity?

Calibration

Are estimated effects systematically too small or too large?

Ranking

Do high-predicted-benefit groups show larger treatment effects in independent data?

Support + uncertainty

Are estimates based on adequate treatment overlap and enough information?

Evaluate the CATE model, not only its ability to predict the observed outcome.

Why In-Sample HTE Patterns Can Mislead

Discover + evaluate on the same data

The model can search for spurious groups.

Examining the same observations again may make those groups look more convincing.

Use independent evidence

Train on one set and evaluate HTE on held-out data.

Cross-fitting can use each observation for evaluation without fitting its prediction on itself.

Exploratory subgroup discovery should be distinguished from confirmatory evidence.

Cross-Fitting: The Basic Workflow

Fold A

Fit models using data outside Fold A.

Predict CATE for Fold A.

Fold B

Fit models using data outside Fold B.

Predict CATE for Fold B.

Repeat

Every evaluated person receives a prediction from a model that did not train on their outcome.

Cross-fitting limits overfitting; it does not fix unmeasured confounding.

CATE Calibration: Does the Magnitude Make Sense?

Example

Model predicts effects near −2 for low-risk patients and −10 for high-risk patients.

In independent evaluation data, do estimated treatment contrasts align with those magnitudes?

What calibration checks

Whether stronger predicted effects correspond to stronger causal contrasts.

Needs an appropriate causal estimator and uncertainty, not raw outcome comparisons.

CATE calibration ≠ ordinary outcome-prediction calibration.

Ranking by Predicted Benefit

First

From training data, learn $\hat{\tau}(x)$.

Second

In held-out data, group units into low, medium, and high predicted benefit.

Third

Estimate each group's causal effect, with adjustment appropriate to the design.

Do not compare untreated outcomes with treated outcomes without a credible design.

Local Overlap: Some CATEs Are Harder to Learn

$$e(x) = P(T = 1 \mid X = x)$$

At $e(x) \approx 0.5$

Treated and control observations are both reasonably represented nearby.

At $e(x) \approx 0.01$ or 0.99

One treatment arm is rare; local counterfactual comparisons are weakly supported.

A forest cannot recover an unobserved treatment arm from modeling alone.

Uncertainty: Estimated CATE Not an Individual Truth

Report

- Predicted treatment effects at relevant X
- Appropriate standard errors / intervals
- Uncertainty in subgroup differences
- Sample size and overlap diagnostics

Avoid

- Treating each $\hat{\tau}(x)$ as a known individual effect
- Calling a visually large difference significant
- Extrapolating to unsupported X regions
- Treating noisy rankings as clinical fact

If intervals are wide, say the data are not informative enough about heterogeneity.

R: Useful Causal-Forest Diagnostics

R code

```
# Calibration diagnostic within grf
test_calibration(cf)

# Forest's estimated average treatment effect
average_treatment_effect(cf)

# Treatment propensity estimated by forest
summary(cf$W.hat)

# Inspect CATE predictions
quantile(predict(cf)$predictions,
         probs = c(.05, .25, .5, .75, .95))
```

Important qualification

test_calibration() is a model diagnostic, not external validation.

For a real HTE validation exercise, use independent/held-out comparisons and appropriate inference.

What Should Responsible HTE Results Include?

Design first

- Population and estimand
- Treatment/outcome definitions
- Measured confounders
- Overlap assumptions

Method + diagnostics

- Model specification
- Sample splitting / honesty
- CATE plots and intervals
- Calibration and stability

Interpretation

- Magnitude and direction
- Clinically/policy meaningful strata
- Exploratory vs confirmatory
- Unmeasured confounding caveat

Explain who benefits, how much, and how confidently—not just which algorithm won.

Final HTE Takeaways and the Next Causal Challenge

What we can now do

- Model treatment × X interactions
- Estimate CATE with S/T/X learners
- Estimate flexible CATE with causal forests
- Evaluate heterogeneity rather than assuming it

What remains unresolved

All of these approaches require a defensible identification strategy.

What if treatment assignment depends on an important confounder U that was never measured?

Next: Unobserved Confounding and Sensitivity Analysis